Module/Pathway Review: Glucarate and Galactarate Catabolism to 2-Oxoglutarate in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Bucket:** KEGG ppu00053 "Ascorbate and aldarate metabolism" (module area: other_kegg_pathway) **Scope reviewed:** three-stage aldarate route: substrate-specific aldarate dehydration → common intermediate 5-dehydro-4-deoxy-D-glucarate → 2,5-dioxopentanoate → 2-oxoglutarate.

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1. Executive summary

The module is **fully satisfiable** in *P. putida* KT2440, and the organism uses the **2-oxoglutarate–forming route** (not the *E. coli*-type aldolytic route to pyruvate + glycerate). All three generic steps are encoded:

- **Substrate entry (D-glucarate):** `gudD` / **PP_4757** (glucarate dehydratase, EC 4.2.1.40) — **covered**.
- **Substrate entry (D-galactarate):** `garD` / **PP_3601** (galactarate dehydratase, EC 4.2.1.42) — **covered**.
- **Common-intermediate dehydration: PP_3599** (5-dehydro-4-deoxyglucarate dehydratase / KDGDH, EC 4.2.1.41) — **covered** (Swiss-Prot reviewed).
- **Terminal oxidation to 2-oxoglutarate: PP_3602** (α-ketoglutarate-semialdehyde dehydrogenase / KGSADH, EC 1.2.1.26) — **covered** (operon-embedded; the correct one of three paralogs).

Two candidate genes are **not** part of this module: **PP_2926** (`udg`, UDP-glucose 6-dehydrogenase) is an anabolic nucleotide-sugar enzyme (mark **not_expected_in_target_taxon**), and **PP_1256** / **PP_2585** are EC 1.2.1.26 paralogs serving other pathways (hydroxyproline and polyamine catabolism). **PP_1171** (`udh`, uronate dehydrogenase) is a legitimate but *upstream* feeder (uronate → aldarate), not one of the three core steps. A genome-wide scan finds **no GarL aldolase** (EC 4.1.2.20), confirming the intermediate can only proceed to 2-oxoglutarate.

Confidence: **high** for satisfiability and gene assignments (direct genome context + Swiss-Prot + one direct KT2440 enzyme characterization). The main residual uncertainty is the *level* of direct wet-lab evidence for growth on glucarate/galactarate in KT2440 itself.

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2. Target-organism pathway definition

Included process (this module): intracellular catabolism of the C6 aldaric acids **D-glucarate** and **D-galactarate** to the TCA-cycle intermediate **2-oxoglutarate (α-ketoglutarate)**, via:

1. D-glucarate → 5-dehydro-4-deoxy-D-glucarate (GudD, EC 4.2.1.40) and/or D-galactarate → 5-dehydro-4-deoxy-D-glucarate (GarD, EC 4.2.1.42);
2. 5-dehydro-4-deoxy-D-glucarate → 2,5-dioxopentanoate (= α-ketoglutarate semialdehyde) + H_2O (KDGDH, EC 4.2.1.41);
3. 2,5-dioxopentanoate + $NAD(P)^+$ → 2-oxoglutarate (KGSADH, EC 1.2.1.26).

Neighboring pathways/maps to keep separate:

- **KEGG ppu00053 "Ascorbate and aldarate metabolism"** and **ppu00040 "Pentose and glucuronate interconversions"** are broad umbrella maps; both contain enzymes not in this module. The bucket resolution pulled in over-broad members (see §5). **ppu01100 "Metabolic pathways"** is an overview map — ignore for satisfiability.
- The ***E. coli*** **D-glucarate/D-galactarate → D-glycerate pathway** (operons `garD`–`garPLRK`–`gudPD`; regulator `sdaR`; enzymes GarL aldolase EC 4.1.2.20, GarR tartronate-semialdehyde reductase, GarK glycerate kinase) is a **distinct downstream branch** and must not be transferred to KT2440, which lacks GarL (P 10762278).
- **Hydroxyproline degradation** (trans-4-hydroxy-L-proline → α-ketoglutarate semialdehyde → 2-oxoglutarate) and **polyamine/amino-acid catabolism** are separate modules that *share the KGSADH enzyme family* — the source of paralog cross-mapping.

Alternate names: "aldarate/saccharate catabolism"; the terminal enzyme appears as "2,5-dioxovalerate dehydrogenase", "2,5-dioxopentanoate dehydrogenase", or "α-ketoglutaric-semialdehyde dehydrogenase (KGSADH)"; the intermediate "5-dehydro-4-deoxy-D-glucarate" = "5-keto-4-deoxy-D-glucarate"; "2,5-dioxopentanoate" = "2,5-dioxovalerate" = "α-ketoglutarate semialdehyde".

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3. Expected step model (target-aware)

#	Step	Reaction (EC)	KT2440 gene	Status
1a	D-glucarate entry	glucarate dehydratase (4.2.1.40)	`gudD` PP_4757	**covered**
1b	D-galactarate entry	galactarate dehydratase (4.2.1.42)	`garD` PP_3601	**covered**
2	common intermediate → 2,5-dioxopentanoate	5-dehydro-4-deoxyglucarate dehydratase (4.2.1.41)	PP_3599	**covered**
3	terminal oxidation → 2-oxoglutarate	KGSADH / 2,5-dioxovalerate dehydrogenase (1.2.1.26)	PP_3602	**covered**
(+)	upstream feed: uronate → aldarate	uronate dehydrogenase (1.1.1.203)	`udh` PP_1171	present (out-of-module feeder)
(+)	transport	MFS glucarate/aldarate transporters	PP_3600, PP_4758	present (accessory)
(+)	regulation	GntR-family regulators	PP_3603, PP_4759	present (accessory)

Both substrate-entry reactions are present, so the module's "either or both entry" option resolves to **both** in KT2440.

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4. Candidate genes and evidence

High-confidence, in-module:

- **PP_4757 (`gudD`, glucarate dehydratase, EC 4.2.1.40, K01706).** Enolase-superfamily D-glucarate dehydratase subgroup. Forms its own genomic cluster with a dedicated MFS transporter (PP_4758), a GntR-family regulator (PP_4759), a Zn-alcohol dehydrogenase (PP_4760) and a HAD hydrolase (PP_4761) — a self-contained, independently inducible glucarate-utilization unit. Evidence: homology + strong genomic context. UniProt TrEMBL (unreviewed). *Caveat:* substrate specificity within the ENS D-glucarate-dehydratase subgroup can be promiscuous; direct assay not confirmed in KT2440.
- **PP_3601 (`garD`, galactarate dehydratase, EC 4.2.1.42, K01708).** Classic GarD; structurally a **distinct non-enolase fold** (P 31811683), the first enzyme of the galactarate branch. Operon-embedded (PP_3599–PP_3602). Evidence: homology + genomic context. TrEMBL.
- **PP_3599 (KDGDH, 5-dehydro-4-deoxyglucarate dehydratase, EC 4.2.1.41, K01707). Swiss-Prot reviewed**; UniProt pathway "D-glucarate degradation; 2,5-dioxopentanoate from D-glucarate: step 2/2". The committed common-intermediate step; catalyzes the decarboxylation/dehydration to α-ketoglutarate semialdehyde. Highest-confidence in-module gene.
- **PP_3602 (KGSADH, EC 1.2.1.26, K13877).** Terminal oxidation to 2-oxoglutarate. Assignment rests on **guilt-by-operon** (directly downstream of `garD`). TrEMBL. This is the correct paralog for the module (see §5).

Present but out-of-module / accessory:

- **PP_1171 (`udh`, uronate dehydrogenase, EC 1.1.1.203, K18981). Swiss-Prot reviewed**; oxidizes β-D-glucuronate → D-glucarate and β-D-galacturonate → D-galactarate. **Directly characterized in KT2440** (cloned and assayed; P 19060141) — the strongest experimental evidence in the candidate set. It *feeds* substrates into the module from hexuronate/pectin catabolism but is not one of the three core dehydration/oxidation steps. Keep it as an "entry expansion," not a core step.

- **PP_3600, PP_4758 (MFS transporters, K03535 glucarate transporter family)** and **PP_3603, PP_4759 (GntR-family regulators)** — accessory transport/regulation, not enzymatic steps.

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5. Gaps, ambiguities, and likely over-annotations

- **Three EC 1.2.1.26 paralogs (PP_1256, PP_2585, PP_3602) — paralog ambiguity + over-propagation.** KEGG maps all three identically to ppu00040/ppu00053/ppu00470 by orthology (K13877). Genome context disambiguates them:
- **PP_3602** → aldarate operon → **the module's terminal enzyme (covered)**.
- **PP_1256** → embedded in the **trans-4-hydroxyproline degradation operon** (PP_1255 D-hydroxyproline dehydrogenase K21060; PP_1257 1-pyrroline-4-hydroxy-2-carboxylate deaminase K21062; PP_1258 4-hydroxyproline epimerase K12658). Its KGSADH activity serves **hydroxyproline** catabolism → reassign; **candidate_uncertain / off-module** for aldarate.
- **PP_2585** → unrelated locus near 8-oxoguanine deaminase (PP_2584), putrescine–pyruvate transaminase (PP_2588) and γ-glutamylaminobutanal dehydrogenase (PP_2589) → **polyamine/amino-acid** catabolic context → **candidate_uncertain / off-module**. KGSADH is a **hub aldehyde dehydrogenase** reused by convergent α-ketoglutarate-semialdehyde–producing pathways; the KEGG multi-mapping is expected over-propagation. The three proteins are **divergent paralogs, not recent duplicates** (pairwise amino-acid identity ≈57–69%: PP_1256–PP_2585 57%, PP_1256–PP_3602 57%, PP_2585–PP_3602 69%), consistent with each being retained for a distinct physiological route — so functional identity cannot be assumed uniform and must be assigned by genomic context/experiment.
- **PP_2926 (`udg`, UDP-glucose 6-dehydrogenase, EC 1.1.1.22, K00012) — spurious module inclusion.** This is an **anabolic** enzyme (UDP-glucose → UDP-glucuronate, nucleotide-sugar biosynthesis), with no reaction on glucarate/galactarate/5-dehydro-4-deoxyglucarate/2,5-dioxopentanoate. Its appearance in the ppu00053 bucket is a KEGG umbrella-map artifact. **Mark not_expected_in_target_taxon** for this module.
- **No GarL aldolase (EC 4.1.2.20) in the genome.** Genome-wide KEGG link queries (K01630/K18929) return nothing → the *E. coli* branch to pyruvate + glycerate is **absent**; the 2-oxoglutarate route is the **sole** fate of 5-dehydro-4-deoxy-D-glucarate. This is a strong pathway-boundary constraint.

- **Cofactor specificity of KGSADH (NAD^+ vs $NADP^+$)** is not established for KT2440; the generic module notes "$NADP^+$". Minor; would need enzyme assay.
- **Direct growth phenotype** (KT2440 on glucarate/galactarate as sole carbon source) is inferred from complete pathway + dedicated transporters/regulators, not cited here from a KT2440 growth study.

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6. Module and GO-curation recommendations

Step status calls:

Module step	Call	Gene(s)
D-glucarate entry (GudD)	**covered**	PP_4757
D-galactarate entry (GarD)	**covered**	PP_3601
5-dehydro-4-deoxyglucarate → 2,5-dioxopentanoate (KDGDH)	**covered**	PP_3599
2,5-dioxopentanoate → 2-oxoglutarate (KGSADH)	**covered**	PP_3602

Candidate-list cleanup:

- **Remove from module / not_expected_in_target_taxon:** PP_2926 (`udg`, anabolic).
- **Reassign off-module (candidate_uncertain for this module):** PP_1256 (→ hydroxyproline module), PP_2585 (→ polyamine/amino-acid catabolism).
- **Retain as out-of-module upstream feeder (annotate as entry expansion, not a core step):** PP_1171 (`udh`).
- **Add accessory context (optional):** transporters PP_3600/PP_4758; regulators PP_3603/PP_4759.

Module-boundary note: the generic module's "either or both substrate-entry" is correct and resolves to **both** for KT2440. No module_needs_revision on structure; however, the **bucket→candidate mapping** (kegg:ppu00053 / ppu00040) is too permissive and should be tightened to exclude anabolic K00012 and non-operonic K13877 paralogs. Consider a **separate module document for hydroxyproline→2-oxoglutarate** (which legitimately claims PP_1256 + the K21060/K21062/K12658 set) so the shared KGSADH family does not collide with the aldarate module. GO annotations: PP_3602 → GO:0047533 (2,5-dioxovalerate dehydrogenase) restricted to the aldarate context; avoid propagating the same GO term to PP_1256/PP_2585 without pathway-specific evidence.

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7. Genes to promote to full `fetch-gene` review

1. **PP_3602** (KGSADH) — highest priority; TrEMBL-only, assignment rests on operon context; confirm it (not PP_1256/PP_2585) is the aldarate terminal enzyme and its cofactor preference.
2. **PP_3601** (`garD`) and **PP_4757** (`gudD`) — TrEMBL; confirm substrate specificity (galactarate vs glucarate) and family placement.
3. **PP_1256** — promote in the context of the **hydroxyproline** module to formally reassign it away from aldarate.
4. **PP_1171** (`udh`) — already Swiss-Prot + direct KT2440 data; low priority, but tag explicitly as an out-of-module feeder.

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8. Key references

- Yoon SH, *et al.* "Cloning and characterization of uronate dehydrogenases from two pseudomonads and *Agrobacterium tumefaciens* strain C58." *J Bacteriol* 2009.

  P 19060141

  — Direct KT2440 evidence that `udh`/PP_1171 oxidizes uronates to aldarates.
- Rosas-Lemus M, *et al.* "Structure of galactarate dehydratase, a new fold in an enolase involved in bacterial fitness after antibiotic treatment." *J Biol Chem* 2020.

  P 31811683

  — GarD as the galactarate-entry dehydratase; distinct fold.
- Monterrubio R, *et al.* "A common regulator for the operons encoding the enzymes involved in D-galactarate, D-glucarate, and D-glycerate utilization in *E. coli*." *J Bacteriol* 2000.

  P 10762278

  — Defines the *E. coli* aldolytic branch (GarL/GarR/GarK; regulator `sdaR`) that KT2440 lacks.

- Bearne SL. "The interdigitating loop of the enolase superfamily..." 2017. P 28179138 . — Enolase-superfamily subgroups relevant to GudD and "galactarate dehydratase 2".
- Database evidence: UniProt (Q88NN6/PP_1171 and Q88GW8/PP_3599 are Swiss-Prot reviewed) and KEGG genome context for *P. putida* KT2440 (ppu), used throughout for operon structure and paralog disambiguation.

Evidence provenance summary

- **Direct for KT2440:** `udh`/PP_1171 enzyme characterization (P 19060141); genome organization/operon structure (KEGG ppu); Swiss-Prot curation of PP_3599 and PP_1171.
- **Homology/database inference:** functional identity of PP_4757, PP_3601, PP_3602 (TrEMBL + KO assignment + operon context); GudD/GarD family placement.
- **Cross-organism (transfer noted):** *E. coli* branch/regulator (P 10762278) — used only to *exclude* an alternative; GarD fold (P 31811683 , structure from another organism) — strong for family identity, neutral for KT2440 specificity.
